Introduction

Domestic violence, defined as abusive behavior used by one partner to gain power or control over the other, is a widespread issue within the United States (The US Department of Justice, 2022). According to the CDC's National Intimate Partner and Sexual Violence Survey, about 41% of women experience some form of intimate partner violence in their lifetime (Centers for Disease Control and Prevention, 2022).

In this paper, we propose that changes in median household income will alter reported instances of domestic violence in the United States. Specifically, we propose that an increase in median household income will result in decreased rates of domestic violence. Among U.S. couples, high income households experience fewer reported instances of domestic violence. This can be attributed to the reduction in financial resource strain experienced by high income households, which reduces stress and potential triggers for violence. The implied correlation between income increases and employment also contributes to this disparity. According to employment protection theory, increased income (and therefore employment) results in less time spent at home, reducing exposure to the violent partner for the female and limiting opportunities for assault for the male. Additionally, employment for females offers increased bargaining power via higher income, allowing greater confidence in negotiations with the abusive partner and supplying financial resources with which to exit the relationship (Farmer & Tiefenthaler, 1999).

Domestic violence not only results in substantial negative effects for victims and those around them, but also has a high external cost. Each year, the cost of intimate partner violence surpasses $5.8 billion, with almost $4.1 billion spent on direct medical and mental health services (Centers for Disease Control and Prevention, 2003). Costly government legislation is also frequently implemented in efforts to address underlying causes of domestic violence, such

as the National Domestic Violence Prevention Action Plan Act of 2021 (Wexton, 2021). Healthcare costs associated with abuse are also substantial: Women on Medicaid who make up to or below 200 percent of the FPL are roughly 6 times more likely to have been admitted to the hospital for an assault while pregnant (Aizer, 2010).

Abuse has also been shown to lead to a decrease in productivity rates, furthering economic cost (Wexton, 2021). The detrimental effects of domestic abuse in the home, resulting in struggles with mental and physical health issues, carry over to a poor workplace performance for female victims, resulting in both reduced output and increased risk for job loss. By addressing abuse and violence effectively, the economic cost of abuse can be reduced to create a more supportive and productive work environment. This not only benefits employers and the economy, but also the individual experiencing the abuse.

Literature Review

We intend to estimate the causal effect of increased median household income on reported instances of domestic violence. By examining changes in median household income for several key subgroups, designated by race, age, presence of children within the home, marital status, and female vs. male ownership of home, we will address the nuances each of these factors introduces to the effect income has at reducing violence. Established literature examines the changes in relationship dynamics between abuser and victim caused by household income shifts through these varied lenses.

Traditional feminist theory articulates that interpersonal violence (IPV) is caused by the female partner's economic dependence on the abusive partner. Pollak (2005) examines the role of bargaining power in marriages and suggests that higher wages for women increase their bargaining power in the marriage, allowing them to negotiate with abusive partners and

potentially leave the relationship. The bargaining power of spouses within a marriage, he argues, is influenced by individual earnings and wage rates, which in turn affects the division of household tasks and the allocation of leisure time among partners. Evidence is found that spouses with higher earnings, owing to higher wage rates, possess greater bargaining power. Pollak's findings provide a critical framework for our research question by presenting clear reasoning behind the negative relationship between women's income and rate of domestic violence: that changes in the household income level, whether for the male or female, affect each partner's respective bargaining power.

Cesur (2022) utilizes the 1994 Earned Income Tax Credit bill as a proxy for a change in household income and arrives at a similar conclusion: that increases in household income are accompanied by decreases in domestic violence rate. The EITC bill allotted differences in access to the tax credit on the basis of household size, mainly whether or not the couple had children. The maximum tax credit for couples with children, the treatment group, was ultimately 7 to 8 times that of the control group, those childless. In line with the idea of resource strain as a significant stressor contributing to home violence, the expansion of earned income via the EITC bill reduced financial constraints on the household, thus limiting a potential cause for disputes within the home. The limitations of this methodology, using child-related tax benefits as a mechanism to examine income changes, are its focus on changes in overall household income in relation to the presence of children. This is relevant to our research in that we also wish to examine how the presence of children impacts the stability and long-term dynamic of the abusive partnership. However, we also wish to take into account the tax benefits and changes occurring when the partners are married versus unmarried. Our research will utilize the United States Census Bureau's data to analyze changes in the median household income in one state among

both married households and families with children. This will provide a more accurate estimation of the impact of family revenue on violence rates, building on Cesar's findings about the impact of overall income increases.

The presence of children has been demonstrated to potentially increase domestic violence. This is due to heightened stress levels, feelings of frustration, and a reduced capacity for reconciling conflicts, as well as the additional strain on resources dependents create. Broadly speaking, the outcomes of the previous studies indicate a link between the quantity of offspring and the incidence of domestic violence targeted toward females and minors. The literature demonstrates having additional children has a positive correlation with increased aggression within the homes of abusive partnerships.

Other sources note that the increased employment rate associated with higher income levels means women will spend less time at home, resulting in less exposure to the abusive partner and thus decreased violence. Bowlus and Seitz (2006) present a model that utilizes cross-sectional data including samples of women aged 25-55 with varied marital statuses, with the regression analysis showing that employment minimizes the likelihood of domestic violence. Women can use employment to prevent incidents of abuse in marriage as it supplies them with a sense of empowerment that may embolden them to exit the relationship. Exposure theory posits that the increases in male employment associated with higher income result in less interaction between the male and female, limiting opportunities for abuse.

Finally, existing literature establishes the importance of race in understanding domestic violence because it has been consistently associated with higher rates of domestic violence among certain racial and ethnic groups (Grossman, 2007). Responses to domestic violence are influenced by social, cultural, and historical factors. This holds relevance to our research

question as we wish to examine both how income changes over time intersect with race and ethnicity, as well as the varying degree to which a change in income affects violence rates for different ethnic groups.

Empirical Methodology

a. Data

From the United States Census Bureau, the analysis data cover median income of each year from 2011 to 2015. From the LAPD Calls for Service, the data cover each Domestic Violence reported incident that occurred in the area. The panel data covers 21 zip codes in Los Angeles with twenty-one total observations, as well as the household median income of all five years. More detailed information can be found in the Metadata Guide.

Description of variables

Variable Names	Description
zipcode	Zipcode
year	Year
medincomeHouseholds	Median Income in Households
medincome25to44years	Median Income of Householders ages 25-44
medincome45to64years	Median Income of Householders ages 45-64
medincomeAsian	Median Income of Asian Householders
medincomeBlack	Median Income of Black Householders
medincomeFamilies	Median Income of Family Identified Households
numDV	Total Number of Domestic Violence
blackratio	Median Income of Black Householders / all Households
asianratio	Median Income of Asian Householders / all Households
youngageratio	Median Income of ages 25-44 / all Ages
oldageratio	Median Income of ages 45-64 / all Ages
familyratio	Median Income of Family Households / all Households

Descriptive Statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
zipcode	105	90513.429	633.991	90003	91601
medincomeHouseholds	105	43984.933	15394.501	12015	73478
medincome25to44years	103	48886.427	17229.751	10960	86593
medincome45to64years	103	38951.961	17766.524	10327	79453
medincomeAsian	40	49326.025	24100.62	11738	86927
medincomeBlack	105	34075.876	17066.931	5147	93537
medincomeFamilies	42	48995.905	18238.556	19819	91831
year	105	2013	1.421	2011	2015
numDV	105	3086.714	1496.454	1026	7369
blackratio	105	.756	.187	.428	1.498
asianratio	40	1.072	.244	.495	1.446
youngageratio	103	1.126	.299	.64	3.146
oldageratio	103	.876	.243	.366	1.28
familyratio	42	1.179	.488	.919	3.442

Fixed Effects Regression

Regression results

numDV	Coef.	St.Err.	t-value	p-value	[95% Conf	Interval]	Sig
blackratio	-201.26	523.409	-0.38	.706	-1323.86	921.341	
asianratio	-82.687	234.37	-0.35	.729	-585.361	419.987	
youngageratio	528.937	717.211	0.74	.473	-1009.327	2067.202	
oldageratio	-508.747	836.534	-0.61	.553	-2302.934	1285.44	
familyratio	476.234	546.306	0.87	.398	-695.477	1647.944	
medincomeHouseholds	-.029	.053	-0.55	.593	-.142	.084	
Constant	3842.095	2872.439	1.34	.202	-2318.675	10002.864	

Mean dependent var	2920.475	SD dependent var	1343.014
R-squared	0.118	Number of obs	40
F-test	0.312	Prob > F	0.995
Akaike crit. (AIC)	455.613	Bayesian crit. (BIC)	467.435

**** p<.01, ** p<.05, * p<.1*

b. Model Specification and Estimation Strategy

To assess the effect of changes in median household income on domestic violence rates, we produced the following regression:

$$DVCalls = \beta_0 + \beta_1\, blackratio + \beta_2\, asianratio + \beta_3\, youngageratio + \beta_4\, oldageratio + \beta_5\, familyratio + u$$

The dependent variable is the number of police dispatch calls related to domestic or family violence taking place each year. These calls, with each call representing one instance of police being sent to address a serious complaint/crime relating to domestic disputes, serve as our proxy measure of domestic violence instances. Our primary variable of interest is median household income, which includes all households regardless of race and size. To allow for the analysis of median income changes across various subgroups, while taking into account issues of multicollinearity, we constructed several median income ratios. The numerator of this ratio contains the subgroup of interest, with the denominator being the median household income for everyone in the zip code of interest. We also included the median income for all households as a method of comparison. The interpretation of our ratio coefficients can be measured as the change in Y, DV calls, associated with a one-unit change in X1, the specified sub-group, for a fixed value of X2, median household income.

We first generated 2 subgroups for the races Black and Asian, examining the ratio for median income for Black and Asian households against all households in the zip code. We anticipate a negative sign for both households, as non-white races have been found to experience higher rates of domestic violence, so increases in income would have greater effects in reducing violence rates than among non-white households. A one-unit increase, i.e., an increase of the total median household income of Black or Asian median income, while keeping the median household income constant, results in a decrease in DV calls.

We then generated a ratio for 2 specified age groups: youngageratio representing ages 25-44, and oldageratio representing ages 45-64. We isolated these age groups in the interest of examining how broad changes in income across the ages most likely to contain abusive partnerships with active, continuous abuse would affect the rate of DV calls. We anticipate both

age groups to have a negative correlation with domestic violence calls. We created a family ratio, comparing the median income of families to overall. Family is anticipated to have negative sign

Results

An increase of the total median household income among Blacks, while keeping overall median household income constant (this is a one-unit increase), results in a decrease in DV calls by 593.724. Similarly, a one-unit increase of the total median household income in the Asian median income, while keeping the median household income constant (this is a one-unit increase), results in a decrease in DV calls by 147.198. Violence is already an issue in racial communities, each change in income level is felt more profoundly impact on those communities, plays a larger role on the DV calls

The coefficient on the family variable is a positive value indicating an increase of 601.335 to DV calls indicating an increase of total median household income among families, while keeping overall median household income constant results. The family variable accounts for households with children, both their own and other dependents, as well as families with married partners. This is not in line with what we expected based on existing literature. We hypothesized that the presence of children would lead to changes in the dynamic and stressors within the family, as well as result in financial strain. Additionally, families with children are likely to receive different tax benefits, meaning that a change in income would have a greater impact on them. However, our results demonstrated that an increase in the median income resulted in an increase in DV calls for that zipcode. This likely indicates a misspecification or error in the construction of our model.

We then generated a ratio for 2 specified age groups: youngageratio representing ages 25-44, and oldageratio representing ages 45-64. We isolated these age groups in the interest of

examining how broad changes in income across the ages most likely to contain abusive partnerships with active, continuous abuse would affect the rate of DV calls. The youngageratio shows that one unit increase causes a 263.212 unit increase in DV calls, while the oldageratio shows a one unit increase causes a 853.863 unit decrease in DV calls. This suggests that areas with a higher proportion of young people may be at greater risk of experiencing domestic violence incidents or may be more likely to report incidents when they occur.

Finally, we included the value for overall median household income in the zip code, expecting that an increase in median income for all households would yield a decrease in DV calls. Our regression was in line with this idea, showing that a one thousand dollar increase in overall median income would cause a decrease in domestic calls by a decrease in DV calls by 750. This is in line with established literature that as the wealth of each household increases, the reduction in resource strain results in fewer instances of domestic violence.

There are many clear shortcomings in our model and data. None of the variables we identified were significant in isolation, and each ratio also possessed wide confidence intervals. Our model contains significant errors stemming from poor construction, bias, and a lack of useful, relevant data to accurately answer our proposed research question. Our attempts to test for joint significance for both race related ratios, Asian and Black, as well as both age group ratios, also yielded insignificance.

The limitations in our analysis stem from our lack of sufficiently detailed crime data. The police dispatch data contains only 2 variables for each observation: the zipcode of where the crime occurred, and the nature of the crime. We isolated for calls related to domestic violence and family violence. However, there is no information on the race, family/household size, marital status, education levels, or employment status of the individual(s) who reported the crime.

Without such specifications, we are unable to directly control for variables such as race, education, household size, and employment/work experience that clearly influence the income levels for each household, as well as the level of violent crime likely to occur. This is a severe limitation to our analysis which we attempted to address those factors indirectly through the creation of median income ratios. We attempted to incorporate these variables into our model via the creation of median income ratios on the basis of subgroups. However, these produced unexpected signs on several of our coefficients that are not in line with basic established literature, and yielded no significant estimators.

Our dataset is also limited by its size. We possess the median household income data for only 21 zip codes in California. We have limited knowledge of broader macroeconomics trends taking place in those counties, such as inflation rates and potential policy changes relating to minimum wage, tax benefits, and other welfare programs aimed at addressing domestic violence. Any one of these factors, especially those related to welfare policy, may be contributing both to changes in income levels in those zip codes and rates of domestic violence calls in those zip codes. We also possess limited knowledge of trends in violent crime taking place in these zip codes. As a result, changes in domestic violence calls may be attributed not just to income changes, but also to broader trends in crime rates, which may vary as a result of any number of other factors affecting all types of crime, not just domestic violence.

An ideal dataset would have allowed us to examine the median income changes for households of a specific race, education level, marital status, and size across zip codes that have not passed significant legislation or welfare programs aimed at addressing domestic violence for a period of at least 10 or more years. This would have allowed us to control for the established correlation between race and both income levels and domestic violence rates, as previous

literature establishes interpersonal violence to be a more pressing issue among communities of color. We would be able to account for the changes to household dynamics that occur with the presence of children, as well as the varied tax and welfare benefits married couples and those with dependents receive. All of these are confounding factors that would influence income, while controlling for policy changes would have allowed us to isolate changes in domestic violence trends occurring from political factors versus income changes.

Discussion & Conclusion

The findings of this study suggest that increasing household income can significantly reduce the incidence of household domestic violence, especially among racial and family size subgroups that have already been identified as being more likely to experience domestic violence. This has important policy implications for governments and organizations working towards reducing violence in households. Policymakers should consider implementing measures to improve the economic situation of families as a key strategy in reducing domestic violence. Such measures could include increasing access to education, raising the minimum wage, and stimulus checks. Increasing access to education, which can improve employment opportunities and earnings potential. Education can also provide individuals with the necessary skills to pursue various career opportunities. Another potential policy intervention is to raise the minimum wage, which would increase the earnings of low-wage workers and reduce their risk of encountering financial strain in the long term. However, in the short term stimulus checks could also help alleviate economic stress for families as a direct and immediate method to increase income. These policies could be targeted towards families that are lower in income.

Future directions for research could include collecting new data that could allow for an easier analysis of the effects of income on domestic violence. Future studies could focus on

different cultures, since cultural differences could lead to differences in how an increase in income impacts domestic violence. Additionally, future research could focus on different sexual orientations, as current literature mostly focuses on heterosexual couples.

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